Research Summary

[**Note:** As the research is currently under review, detailed results, specific statistical estimates, and other sensitive findings have been intentionally omitted from this summary. The document focuses on the research objectives, analytical framework, methodology, and overall contribution.]

Socioeconomic and Health-System Correlates of Cesarean Delivery in Bangladesh

This research examines the socioeconomic, maternal, reproductive, healthcare, geographic, and delivery-context factors associated with cesarean delivery in Bangladesh using nationally representative Bangladesh Demographic and Health Survey (BDHS) 2022 data.

The study develops an integrated analytical framework that combines **survey-weighted statistical inference, causal-inference-oriented robustness analysis, machine learning, Explainable AI, and model-based policy simulation.**

What Was Done

The analysis began with data cleaning, variable preparation, exploratory analysis, and survey-weighted estimation to account for the complex sampling design of **BDHS**. Associations between potential determinants and cesarean delivery were then examined using Generalized Estimating Equations (GEE).

To assess the robustness of important associations, the study incorporated propensity-score-based inverse probability of treatment weighting **(IPTW),** covariate-adjusted weighted **GEE,** and E-value sensitivity analysis. These approaches were used to evaluate the stability of observed relationships while maintaining a careful distinction between association and causal interpretation.

A comprehensive machine-learning framework was subsequently developed using multiple classification algorithms, including Logistic Regression, **SVM**, **Random Forest**, **XGBoost**, **AdaBoost, Extra Trees, and a stacking ensemble**. Model validation, hyperparameter optimization, and performance assessment were incorporated into the modeling workflow.

To make the predictive models interpretable, **SHAP** (SHapley Additive exPlanations) was used to identify influential predictors and understand their contributions to individual predictions. Selected nonlinear dependencies and feature interactions were further explored using partial-dependence-based interaction analysis.

Finally, the study incorporated model-based counterfactual policy simulations. Predefined changes in selected socioeconomic, healthcare-utilization, demographic, and delivery-context variables were applied to the fitted model to examine how predicted cesarean-delivery probabilities responded under alternative scenarios. These simulations were treated as predictive sensitivity analyses rather than causal estimates of policy interventions.

Research Contribution

The key contribution of this work is the integration of population-representative statistical analysis with predictive machine learning and Explainable AI, while explicitly separating statistical association, predictive behavior, and model-based scenario analysis.

The framework provides a data-driven perspective on cesarean delivery in Bangladesh and highlights how socioeconomic conditions, healthcare utilization, and delivery settings interact within the observed data. It also demonstrates how interpretable machine learning and scenario analysis can complement conventional statistical approaches in population-health research.

Research Framework

Data Preparation → Survey-Weighted Analysis → GEE Modeling → IPTW & Sensitivity Analysis → Machine Learning → Stacking Ensemble → SHAP & Interaction Analysis → Policy Scenario Simulation

Future Direction

The findings provide a foundation for further research integrating richer clinical, provider, and facility-level information to better distinguish medically indicated cesarean delivery from potentially avoidable use and to develop more robust, externally validated approaches for maternal-health research.